Milrinone

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Continuing Education Activity

Milrinone is a medication indicated for cardiac support in patients with acute heart failure, pulmonary hypertension, or chronic heart failure. It functions by improving cardiac contractility (inotropy), cardiac relaxation (lusitropy), and inducing vasodilation and has the overall effect of increased cardiac output, improvement of left ventricle-arterial coupling, and enhanced cardiac mechanical efficiency. Its use is primarily in the perioperative and ICU settings, although it also has utility for outpatient therapy in select patient populations. This activity will review its mechanism of action, indications, and the potential harm and benefits associated with its use. It will also discuss its role throughout various specialties of medicine, including ICU care, perioperative care, use in pediatric populations, and the now discontinued use in the outpatient setting as an oral medication.

Objectives:

- Describe the mechanism of action of milrinone.
- Identify conditions that indicate milrinone therapy.
- Summarize the potential adverse effects associated with milrinone.
- Explain why the use of milrinone requires thoughtful planning and discussion throughout the interprofessional team with other professionals and specialists involved in the patient's perioperative, ICU, and outpatient care.

Access free multiple choice questions on this topic.

Indications

Milrinone is a medication indicated for cardiac support in patients with acute heart failure, pulmonary hypertension, or chronic heart failure. It functions by improving cardiac contractility (inotropy), cardiac relaxation (lusitropy), and inducing vasodilation and has the overall effect of increased cardiac output, improvement of left ventricle-arterial coupling, and enhanced cardiac mechanical efficiency. Its use is primarily in the perioperative and ICU settings, although it also has utility for outpatient therapy in select patient populations.[1]

FDA Approved Indication

Milrinone is approved for short short-term IV therapy for patients with acute decompensated heart failure with reduced ejection fraction in need of inotropic support.[2]

Use in the Perioperative Setting

Milrinone is often used during various cardiac surgeries, including coronary artery bypass graft surgery, cardiac transplantation, and other cardiac surgeries that require cardiac support. Likewise, it has utility in non-cardiac surgeries for patients with acute decompensated left ventricular heart failure, acute right ventricular heart failure, or pulmonary artery hypertension.

Cardiac Units and the ICU

Milrinone is used mostly in the ICU and the cardiac unit for cardiac support in patients with acute heart failure, weaning patients with pre-existing left ventricular dysfunction from cardiopulmonary bypass, or as a temporizing agent for patients with plans to undergo cardiac surgery or transplantation. In addition, in the neonate population, it is indicated to treat persistent pulmonary hypertension (i.e., neonates with a congenital diaphragmatic hernia).[3][4]

Outpatient Use of Milrinone

Clinicians direct the use of milrinone in the outpatient setting to patients who have severe symptoms of congestive heart failure (CHF) refractory to optimal medical therapy. Previously, an oral version was in use in the outpatient setting for symptomatic treatment of New York Heart Association (NYHA) class III/IV CHF; however, this fell out of favor due to increased patient mortality secondary to ventricular arrhythmia and sudden cardiac death.[5] (see Toxicity section)

In pediatric patients with congenital heart failure, outpatient milrinone infusion regimens are a means of bridging patients until they can undergo cardiac transplantation, initiate mechanical circulatory support, or it may work palliatively in those that are not eligible for transplant/mechanical intervention. This method of treatment is effective for improving patient symptoms and decreasing the number of hospitalizations.[6]

Mechanism of Action

This section discusses the pharmacology of milrinone.

Phosphodiesterase Inhibition

Milrinone is the phosphodiesterase inhibitor drug class. Phosphodiesterase is an enzyme that hydrolyzes the second messenger cyclic adenosine monophosphate(cAMP) and guanosine monophosphate(cGMP), terminating their effects in various tissues. There are several phosphodiesterase enzymes throughout the body. Phosphodiesterase III is present in the cardiac sarcoplasmic reticulum, smooth muscle in arteries and veins. Milrinone is selective for phosphodiesterase III at low doses and nonselective at high doses.[7]

Cardiac Effects of Milrinone

In the myocardium, PDE III inhibition leads to increased contractility (inotropy) and improved relaxation (lusitropy). This effect leads to improved systolic and diastolic function and optimizes the cardiac output. Increased heart rate (chronotropy) also occurs but is less pronounced than the increases in heart rate seen with medications in the catecholamine class. Inhibition of phosphodiesterase III prevents cAMP breakdown, increasing protein kinase A activity, leading to phosphorylation of calcium ion channels in the sarcoplasmic reticulum and increasing calcium availability myocyte sarcomere. The aforementioned increased calcium availability manifests in increased cardiac inotropy and chronotropy. Consequently, PDE III inhibition by milrinone causes increased calcium reuptake into the sarcoplasmic reticulum, resulting in enhanced myocardial relaxation (lusitropy) with improved diastolic function.[8][9]

Vasoactive Effects of Milrinone

In the vasculature, PDE III inhibition prevents cGMP metabolism in the smooth musculature and results in vasodilation in both arteries and veins. The vasodilatory effects seen with milrinone are more potent than beta-2 agonists, including dobutamine and isoproterenol. Milrinone is available in an inhalational formula for directed vasodilation of the pulmonary vasculature to treat pulmonary hypertension.[10][11]

Administration

Dosing options for milrinone include:

Intravenous Administration

- Loading doses: 25 to 50 mcg/kg (loading dose given over 10 minutes)
- Infusion rates: ranging between 0.375 and 0.75 mcg/kg/min[12]

Inhalation Administration

- Not an FDA approved use; there is no consensus on the dosing via this route[13]

Pediatric Home Administration

- Infusion rate: 0.3mcg/kg/min to 1mcg/kg/min[14]

Pharmacokinetics

- T1/2: 2 to 2.5 hours
- Volume of distribution: 0.13 to 0.14 L/kg/hr
- Elimination: via urine, renal clearance 0.3 L/min (90% recovered in urine in 8 hours)[15]

Adverse Effects

The most feared adverse effect of milrinone is its potential to induce hemodynamic changes and arrhythmias. Milrinone may cause ventricular tachyarrhythmia, which may lead to cardiac ischemia or sudden cardiac death. These changes are not shown to follow a dose-dependent relationship. Milrinone can cause an increase in venous vessel capacitance, leading to decreased preload and manifesting as headaches, syncope, and severe hypotension. Unlike tachyarrhythmias, hypotension occurs in a dose-dependent relationship.[15]

Anagrelide may enhance the toxic effect of Milrinone. Both milrinone and anagrelide inhibit the phosphodiesterase III enzymes. Therefore anagrelide prescribing information states that clinicians should avoid concomitant use of milrinone with anagrelide.[16]

Besides its hemodynamic and arrhythmogenic effects, milrinone may also affect platelet function and inflammatory pathways. It may block platelet aggregation, suppress neointimal hyperplasia associated with endothelial injury, and attenuate the proinflammatory effects of cardiopulmonary bypass.[17]

Contraindications

Milrinone is contraindicated in patients with hypersensitivity to any of its components. It is relatively contraindicated in patients with severe heart failure or severe pulmonary hypertension. In severe pulmonary hypertension, generalized vasodilation of pulmonary vasculature may worsen V/Q mismatch and lead to worsened hypoxemia. When considering the use of milrinone in these populations, it is advisable to consult a specialist for expert guidance.[15][18]

The use of milrinone is generally contraindicated in patients with acute kidney injury and end-stage renal disease, as it primarily undergoes renal excretion. Hence, Clinicians should reduce the infusion rate in patients suffering from renal impairment.[19]

Monitoring

Milrinone is primarily for use in the ICU and perioperative setting. Before initiating this medication, a right heart catheterization may be considered for obtaining hemodynamic measurements to establish the patient's baseline parameters and gauge the patient's response to continuous infusion. Repeat or dosing monitoring is not routine due to the risks associated with repeat vascular access or continuous indwelling catheters. Pulmonary artery catheterization for monitoring pulmonary pressures should be done with discretion and only after considering a risk-benefit analysis case-by-case basis.[14]

Serum milrinone levels are not routinely necessary as the arrhythmogenic toxic effects have not shown a dose-dependent linear relationship. Outpatient used of milrinone is routinely not recommended due to increased patient mortality secondary to ventricular arrhythmia and sudden cardiac death.[5][15]

Toxicity

Cardiovascular toxicity is primarily seen in patients receiving milrinone chronically and manifests as tachyarrhythmias and sudden cardiac death. At high dosing, patients may also experience hypotension and syncope.[14][20] Milrinone had a previous use as an oral formulation for outpatient use in patients with NYHA class III and IV chronic heart failure (CHF) to improve symptoms and decrease the frequency of hospital admissions. However, this practice was discontinued following the PROMISE trial in 1991 due to safety concerns. This double-blinded clinical trial assessed patients with CHF class III/IV placed on milrinone or placebo and ended early due to increased mortality in the milrinone group secondary to ventricular tachyarrhythmias and sudden cardiac death.[18][5]

Enhancing Healthcare Team Outcomes

Milrinone can be a useful medication for providing cardiac support in patients with acute and chronic heart failure, providing intraoperative cardiac support, and acting as a bridge to definitive surgical or mechanical support or for palliative support in specific patient populations. However, due to this medication's toxicity and the risks associated with invasive monitoring, ICU providers should carefully perform a risk-benefit analysis before initiating patients on this medication. Nurses should be familiar with the dosing of the drug and provide additional monitoring for adverse effects and patient compliance. Pharmacists should work closely with the prescriber, verify dosing, and perform medication reconciliation to preclude any drug-drug interactions. Given the toxicity and potential adverse event profile of milrinone, an interprofessional, collaborative team effort is the best way to drive patient outcomes and prevent adverse events optimally. [Level 5]